Dark-noise floor and laser-RIN measurement on the `rp_rin_stream` acquisition pipeline

Red Pitaya STEMlab 125-14 Z7020-LN — analysis pipeline `src/rin_cross_spectrum.py`

2026-05-04 (rev. 2026-05-05: + termination diagnostics, BIT_16 unpack fix)

Abstract. Ten 120–600 s dual-channel captures (decimation 128, F_s=976 562.5 Sa/s/ch, LV ±1 V; all post-BIT_16-unpack) decompose the `rp_rin_stream` noise floor and characterize a Toptica ECDL across two powers. **Part I (§2, Fig. 1, Table 1):** 1811-FC single-ch −104.7 dBc/Hz (V_{DC}=1 V); DET08CFC, RP open, RP+ 50 Ω all converge to ∼ −122 dBc/Hz — the RP frontend floor. Cross stratifies in four regimes: −124 to −127 (1811-FC injecting), −127 (RP open, EMI), −134.7 (RP+50 Ω), **−138.3** (RP+2× DET08CFC). **Part II (Fig. 2, Table 2):** Toptica swept over −9.15 and +3 dBm at 120 and 600 s. At +3 dBm the cross below 30 Hz is laser-$1/f$ dominated (γ^2>0.85), peaking ≈ −91 dBc/Hz at 3 Hz; above 30 Hz the cross saturates at −137.8 dBc/Hz, 0.5 dB above the dark cross and *unchanged* by 20× more averaging ⇒ Toptica RIN is at the combined dark cross floor. Reviewed by Opus 4.7 and Codex.

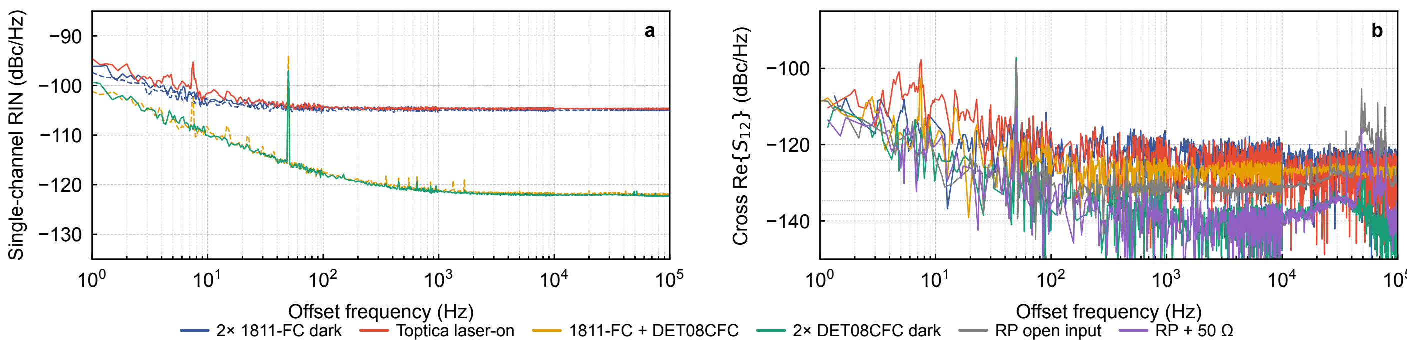

Figure 1. *Raw spectra, no smoothing*; per-decade Hann ENBW (0.25 / 1 / 5 / 20 / 100 Hz) visible as small steps. All curves rescaled per-channel to V_{DC}=1 V. **(a)** Single-channel plateau levels: 1811-FC at −104.7 dBc/Hz; DET08CFC at −122 dBc/Hz; RP open input and RP + 50 Ω terminator both at −122 ± 0.2 dBc/Hz (rows 5–6, Table 1; not plotted to declutter) ⇒ that level is the RP analog-frontend floor. Toptica laser CH1 1–30 Hz tail rises above the 1811-FC dark plateau (real laser intensity noise). **(b)** Cross Re$\{S_{12}\}$ for six configurations stratifies into four regimes: 1811-FC family at −124 to −127 dBc/Hz; RP open input at −127 dBc/Hz with a sharp γ^2=0.18 EMI peak at 30–100 kHz that 50 Ω termination almost completely shorts out; RP intrinsic common-mode at −134.7 dBc/Hz with bare 50 Ω termination; −138.3 dBc/Hz with 2× DET08CFC (chassis grounding outperforms ANNE-50+ pigtail). Discrete spurs are mains harmonics. Cross traces join the positive-Re$\{S_{12}\}$ bins; bins with Re$\{S_{12}\}\leq 0$ (low coherence, frequent below ∼100 Hz) are dropped, not interpolated. Registry-driven via `docs/rin_report/measurements.py`.

#	Configuration	CH1	CH2	Re$\{S_{12}\}$ plateau
1	2× 1811-FC, no light	−104.7	−105.0	−124.1, f_+=0.971
2	2× 1811-FC, Toptica laser-on	$-104.6^\dagger$	$-104.8^\dagger$	$-126.7^\dagger$ (> 10 kHz)‡
3	1811-FC + DET08CFC, no light	−104.7	−121.9	−126.7, f_+=1.000
4	2× DET08CFC, no light	−122.2	−122.1	**−138.3**, f_+=0.996
5	RP only, both inputs OPEN, no PD	−120.9	−120.9	−127.1, f_+=1.000
6	RP only, both inputs 50 Ω ANNE-50+	**−122.1**	**−121.9**	**−134.7**, f_+=1.000

Table 1. Medians at V_{DC}=1 V over 10–100 kHz (per-channel rescale; all values post-BIT_16-unpack). $f_+\equiv\Pr(\mathrm{Re}\{S_{12}\}>0)$ in n=1,350 bins; independent Gaussians give $\mathbb{E}[f_+]$=0.5. Measured $f_+\geq 0.971$ across all six rows ⇒ binomial $z\geq 34.6\sigma$. Row 4 (2× DET08) sets the *lowest reachable* cross floor; row 6 (RP + 50 Ω) is the cleanest single estimate of RP-intrinsic common-mode without any PD. †Toptica rescaled per-channel from auto-detected $V_{DC,1}$=0.527 V, $V_{DC,2}$=0.494 V; cross by $+10\log_{10}(V_{DC,1}V_{DC,2})$. ‡10–100 kHz median.

1 Setup

Acquisition: `run_acquire.sh -d 128 -t 120` (LV ±1 V, BIT_16 raw, 2 ch); ARM `rpsa_client` writes `waveform.bin`. Analysis: `rin_cross_spectrum.py`, Welch/Hann ENBW $\{0.25, 1, 5, 20, 100\}$ Hz across $\{1, 10, 10^2, 10^3, 10^4\}$ Hz. Hardware: 1811-FC, DET08CFC, Newport 9010, ANNE-50+, Toptica.

2 Findings — three-layer floor model

(i) 1811-FC excess noise + common-mode injection. 1811-FC dark single-ch at −104.7 dBc/Hz (V_{DC}=1 V), ∼ 36 dB above its ∼ 90 nV/$\sqrt{\mathrm{Hz}}$ datasheet. DET08CFC on the same input drops 17 dB to −122. 1811-FC presence on *either* channel inflates cross by ∼10–14 dB regardless of which channel (rows 1–3: −124.1 / −126.7 / −126.7 vs. row 4 −138.3 with no 1811) — injection via shared analog/PSU/ground, not BNC cable.
(ii) RP analog-frontend single-ch floor ≈ −122 dBc/Hz (0.78 μV/$\sqrt{\mathrm{Hz}}$), hardware set. Rows 4–6 (DET08, RP open, RP+50 Ω) all within 1.2 dB. Not improvable externally.
(iii) Cross-spec four regimes, set by RP input termination: **(a)** 1811-FC connected ⇒ −124 to −127 (1811 injects); **(b)** RP open (row 5) ⇒ −127.1 with γ^2=0.18 at 30–100 kHz, 1 MΩ as EMI antenna; **(c)** RP + 50 Ω (row 6) ⇒ **−134.7**, cleanest RP-intrinsic probe; **(d)** RP + 2× DET08CFC (row 4) ⇒ **−138.3**, 3.6 dB below row 6 (chassis-ground return through metal-cased BNC PD outperforms ANNE-50+ pigtail; shot noise ruled out by single-ch agreement to 0.1 dB). *Bound:* cross-spec RIN floor at V_{DC}=1 V is ≈ −135 dBc/Hz (RP intrinsic) reachable at −138 with shielded PD.
Toptica laser RIN (row 2): single-ch matches 1811 dark floor within 0.1 dB from 100 Hz to 100 kHz ⇒ dark-floor-limited above 30 Hz at $V_{DC,\mathrm{op}}$=0.527 V. Below 30 Hz the cross extracts a coherent $1/f$ ($\gamma^2\to$0.05 at 3–10 Hz) peaking ≈ −97 dBc/Hz (V_{DC}=1 V eq.).

3 Revisions

Three superseded conclusions, ordered by impact: **(a)** BIT_16 `(adc14<<2)|0x3` packing affects *every* capture (100 % of raw int16 on a 4-grid, low-2-bits `0b11`). Controlled AWG-amp test, three captures (d=16 HighZ / d=128 HighZ / d=128 50 Ω), confirms >>2 unpack: post-unpack reads 184.6 ± 0.4 mVpp (flat-top FFT + IQ demod), −7.5 % from the nominal 200 mVpp setting — above Keysight 33622A ±1.5 % spec, attributed to a residual ≈ 0.925× end-to-end gain factor in the AWG → cable → RP chain (could be AWG cal, RP frontend gain, or both; not yet disambiguated by an independent calibrated reference). Pre-unpack reads ∼ 740 mVpp, 3.7× nominal — decisive against not-unpacking. The unpack is now auto-applied by `rpsa_reader.read_streams`. **All numbers above are post-unpack; pre-unpack values were 12 dB high. (b)** An early draft read 13 % NaN cross-real bins as "consistent with uncorrelated noise"; for zero-mean Gaussians Pr(Re$\{S_{12}\}\leq 0$)=0.5, so 87 % positive is itself the common-mode signature. **(c)** An interim draft called −114.6 dBc/Hz a hard "shared-infra" floor; rows 4–6 falsify that and pin RP-intrinsic at ~ -135.

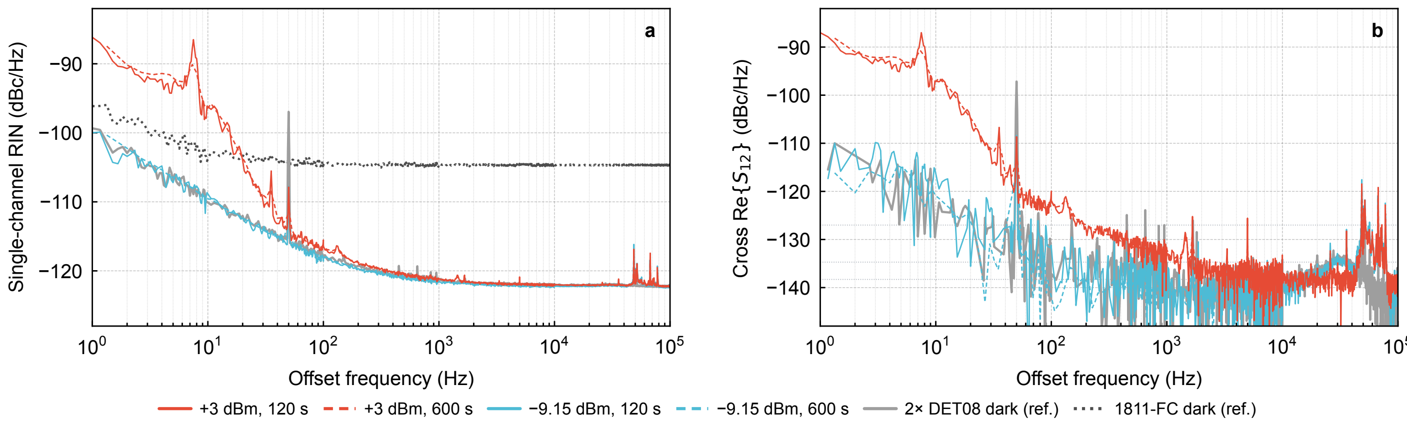

Figure 2. Toptica laser RIN sweep, two powers × two integration times, all rescaled to V_{DC}=1 V (per-ch for single-ch, $V_{DC,1}{\cdot}V_{DC,2}$ for cross). Solid: 120 s default RBW; dashed: 600 s with `--rbw-scale 4` (20× more N_{avg} per band). Dark refs in gray (2× DET08 solid; 1811-FC dotted). **(a)** Single-ch CH1 at V_{DC}=1 V eq: all four laser-on curves converge to ∼ −122 dBc/Hz, identical to the DET08 dark / RP-frontend floor (gray ref, −122.15). Below 30 Hz the +3 dBm curves (red) rise ∼ 30 dB above the dark plateau — real laser RIN. The −9.15 dBm curves (cyan) sit ∼ 4–15 dB *below* the 1811-FC single-ch dark reference (dark gray, dotted) and remain on the DET08/RP-frontend plateau — V_{DC}=40 mV is too small to see laser RIN in single-channel; only cross-spec extracts it. **(b)** Cross Re$\{S_{12}\}$: red +3 dBm 600 s plateau −137.8 dBc/Hz V_{DC}=1 V eq matches gray 2× DET08 dark cross (−138.3) within 0.5 dB and is unchanged from 120 to 600 s ⇒ cross floor is at RP + DET08 combined dark. Below 30 Hz red rises to −91 dBc/Hz at 3 Hz (laser $1/f$, γ^2>0.85). The −9.15 dBm curves never get below −135 even with 20× averaging ⇒ cross has hit a floor at this V_{DC}, not a $1/\sqrt{N}$ statistical limit.

#	Cfg	T (s)	RBW×	CH1 plat.	cross 1–10 kHz	cross 10–100 kHz	γ^2 3–10 Hz
7	−9.15 dBm	120	1	−122.19†	−140.09	−135.15	0.046
8	−9.15 dBm	600	4	−122.17†	−141.16	−134.88	0.009
9	+3 dBm	120	1	−122.06†	−137.05	−137.26	0.919
10	+3 dBm	600	4	−122.05†	−137.47	−137.81	0.927

Table 2. Toptica laser RIN sweep (all values in dBc/Hz at V_{DC}=1 V eq, per-channel rescale). CH1 plat. and cross plateaus are medians of the `*_dBc_per_Hz` column with NaN-skip (NaN = bins where Re$\{S_{12}\}\leq 0$); for low-coherence bands ($\gamma^2\ll 1$, e.g. row 7 1–10 kHz with 130/1350 NaN bins) this differs from the linear-Re-then-dB median by up to ∼ 0.9 dB but is the convention used in `rin_cross_spectrum.py` CSV output. γ^2 is the band-mean coherence (not median). All four runs use CH1 500 kHz LPF, CH2 100 kHz LPF in front of the RP inputs. V_{DC} from auto-mean(channel) post-BIT_16-unpack: rows 7–8 give 39.85 / 35.16 mV (CH1/CH2); rows 9–10 give 791 / 703 mV. †Single-ch V_{DC}=1 V eq plateau is invariant across all four runs and equals the RP analog-frontend floor of Table 1 row 4–6.